La Scuola d'Italia / Dr. Huson / IB Math 12
10 November 2026

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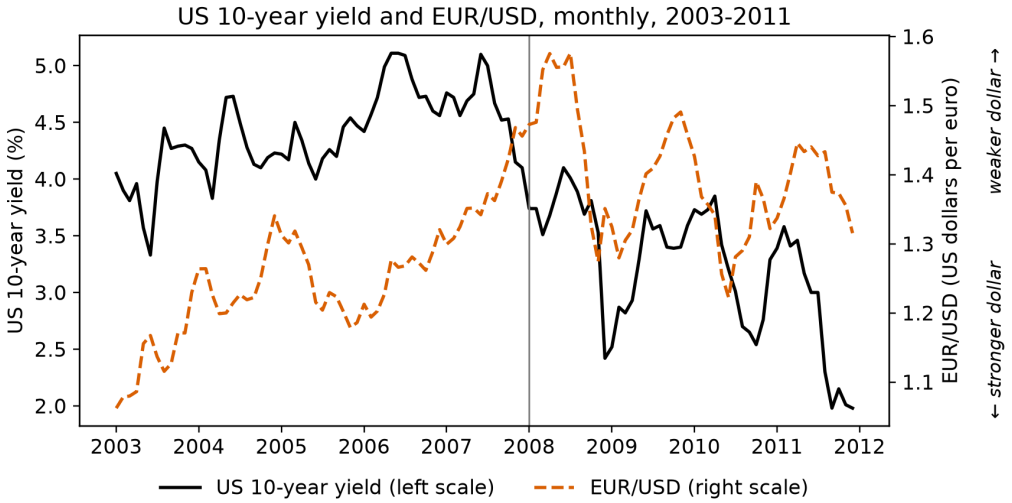
1.12 Classwork: Simpson's Paradox — correlating levels and changes

Use your GDC. Give answers correct to three significant figures. Data: monthly averages, January 2003 to December 2011 (sources: FRED, OECD).

An Italian investor in US bonds. In January 2003 an investor in Milan changes EUR 100 000 into US dollars and buys US 10-year government bonds (Treasuries). The investment carries two risks.

- **Interest-rate risk.** When the US 10-year *yield* rises, the price of bonds already issued falls.
- **Currency risk.** The investor will change the dollars back into euros. The exchange rate EUR/USD is the number of US dollars that one euro buys. A *higher* EUR/USD means a *weaker* dollar.

1. The graph shows the US 10-year yield (solid line, left scale) and EUR/USD (dashed line, right scale) for each month.



Month	Jan 2003	Jul 2008	Nov 2008	Dec 2008
US 10-year yield (%)	4.05	4.01	3.53	2.42
EUR/USD	1.0622	1.5759	1.2744	1.3511

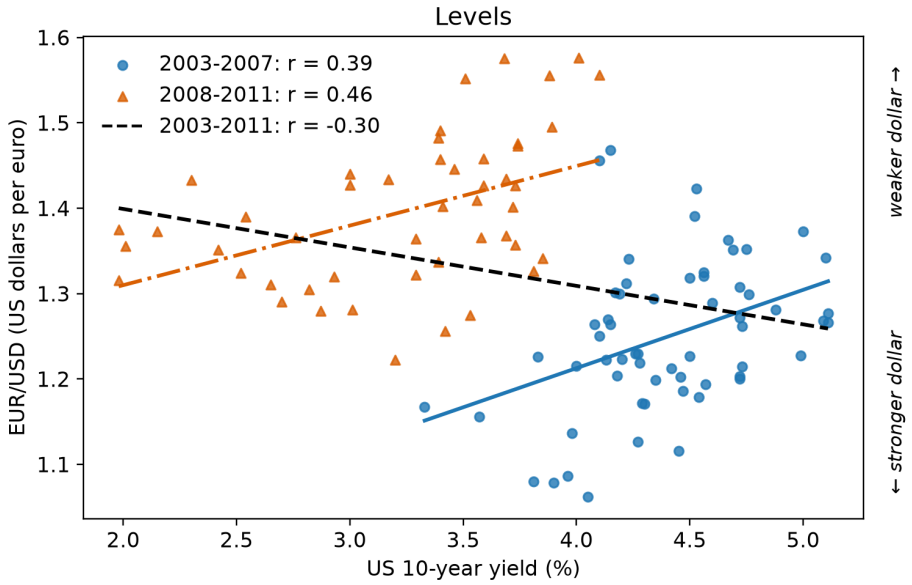
(a) Find the number of US dollars the investor receives for EUR 100 000 in January 2003.

(b) Ignoring interest and any change in the bond price, find the value of these dollars in euros in July 2008. Find the percentage change from the investor's EUR 100 000.

(c) Did the dollar become stronger or weaker between July and November 2008? Find the value in euros of the same dollars in November 2008.

(d) Describe the change in the US 10-year yield between July and December 2008. Explain whether this change was good or bad for the investor.

2. Each point in the scatter diagram is one month: the US 10-year yield, x, and EUR/USD, y. Circles are 2003–2007 and triangles are 2008–2011.



Period	mean yield, $\bar{x}$ (%)	mean EUR/USD, $\bar{y}$
2003–2007 (60 months)	4.40	1.249
2008–2011 (48 months)	3.23	1.396

(a) Write down the value of Pearson's r for 2003–2007, for 2008–2011, and for the whole period 2003–2011. Describe each correlation.

(b) Use the table of means to explain how the correlation can be positive in each period but negative for the two periods together.

(c) Simpson's paradox is a relationship that holds within each group but reverses when the groups are combined. A student concludes from the whole-period value of r: "When US yields rise, the dollar gets stronger." Explain why this conclusion is not justified.

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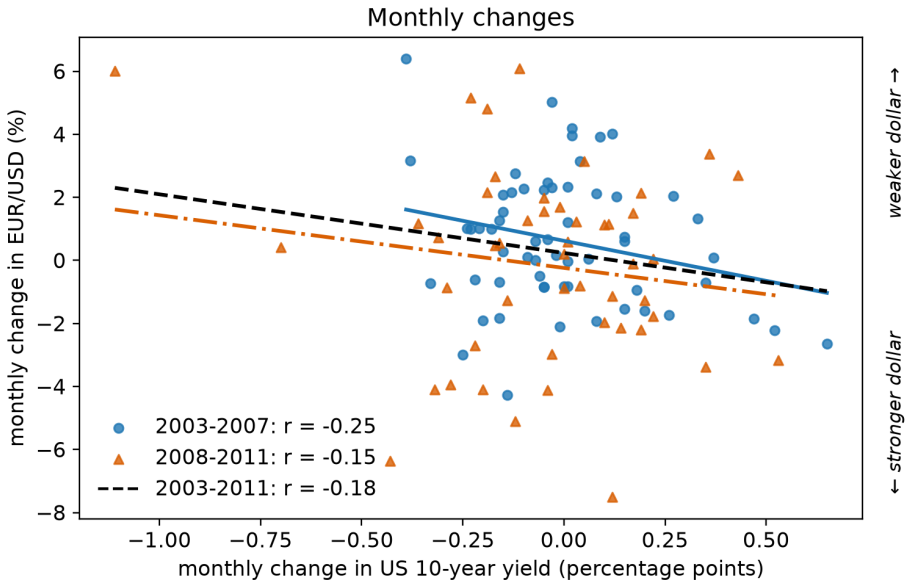
3. Finance compares *changes*, month to month. For the yield the change is in percentage points; for the exchange rate it is the percentage change:

$$\text{change in yield} = x_{\text{this month}} - x_{\text{last month}}, \qquad \text{change in EUR/USD} = \frac{y_{\text{this month}} - y_{\text{last month}}}{y_{\text{last month}}} \times 100.$$

Month (2008)	Jul	Aug	Sep	Oct	Nov	Dec
US 10-year yield (%)	4.01	3.89	3.69	3.81	3.53	2.42
change (percentage points)		−0.12				
EUR/USD	1.5759	1.4955	1.4342	1.3266	1.2744	1.3511
change (%)		−5.10				

(a) Complete the table.

The scatter diagram shows the monthly changes for all 108 months.



(b) Write down the three values of r for the monthly changes. Compare them with your answers to 2(a).

(c) In a month when the US yield rises, the investor's bonds lose value. Use the sign of r to state what tends to happen to EUR/USD in that month, and explain how this affects the investor's total risk.

4. A small data set shows the same effect.

Period	1			2		
x	1	2	3	6	7	8
y	5	4	3	10	9	8

(a) Find r for period 1, for period 2, and for all six points.

(b) Sketch the six points. Explain the result of part (a).

(c) Prices tend to drift up or down over the years, so their average level is different in each period. Monthly changes average close to zero in every period. Explain why the correlation of changes is much less affected by Simpson's paradox than the correlation of levels.

(d) A research paper finds $r = 0.3$ between the prices of gold and oil from 2015 to 2025, but $r < 0$ in each half of the period. Write two sentences of advice to the author.